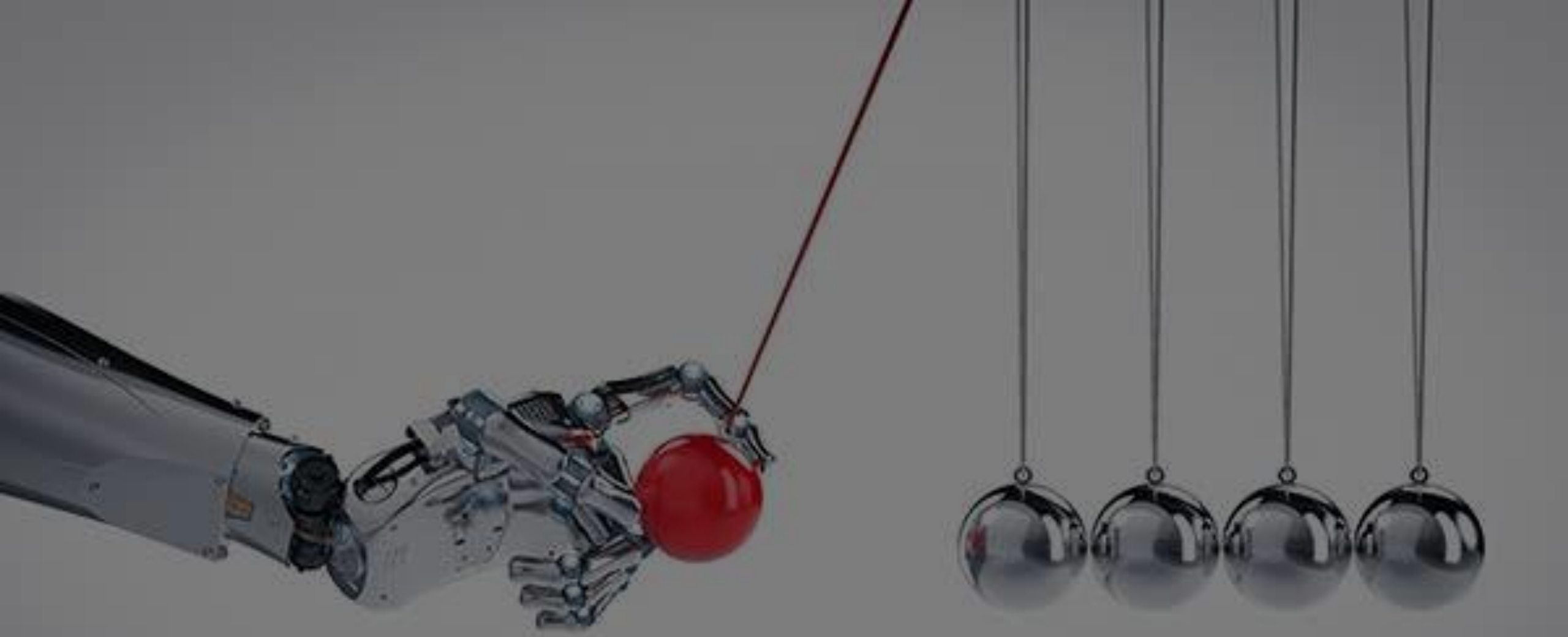
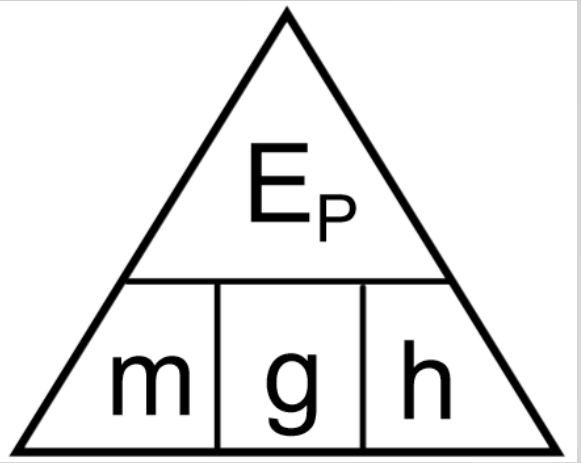


MECHANICAL ENERGY

GRAVITATIONAL POTENTIAL ENERGY

Gravitational potential energy is: *the energy an object possesses due to its position relative to a reference point*

$$E_p = mgh$$



E_p is gravitational potential energy measured in joules (J)
m is mass measured in kilograms (kg)
g is gravitational acceleration (g_{earth} = 9,8 m.s^{-2})
h is height above a reference point measured in metres (m)

Gravitational potential energy depends on the mass of the object and its height above the earth or a specific point.

Gravitational potential energy is a **scalar quantity**.

The point where E_p is nil (ie: where h = 0 m) can be chosen voluntarily and thus does not necessarily have to be the surface of the earth.

***Example**:*

A weightlifter raises weights with a mass of 200 kg from the ground to a height of 1,5 m. By how much does the gravitational potential energy of the weights increase?

***Answer**:*

$E_p = mgh$
$= (200)(9{,}8)(1{,}5)$
$= 2940\ J$



Question 1: Brian has an object suspended in the air. It has a mass of 100 kg and is 25 m above the ground. What is the object's potential energy?

$$E_p = mgh$$
$$= (100)(9,8)(25)$$
$$= 24500 \text{ J}$$

Question 2: The mass of the rock is 1220 kg. It had 5978 J of potential energy before it rolled down the hill. Calculate the height of the hill.

$$E_p = mgh$$
$$5978 = (1220)(9,8).h$$
$$h = 0,5 \text{ m}$$

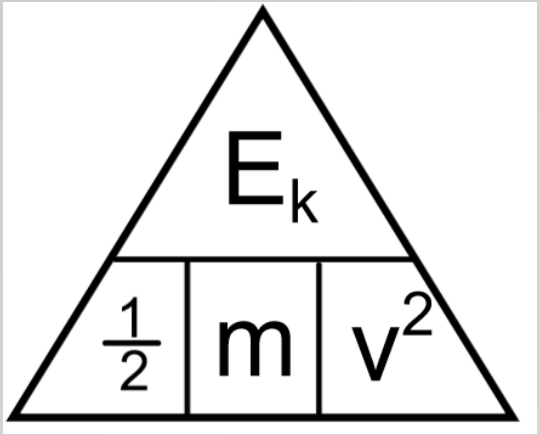
Question 3: Maria is holding a ball at a height of 2 m. The ball has a potential energy of 12,15 J, calculate the mass of the ball.

$$E_p = mgh$$
$$12,15 = m.(9,8)(2)$$
$$m = 0,62 \text{ kg}$$

KINETIC ENERGY

Kinetic energy is: *the energy an object has as a result of the object's motion*.

$$E_k = \frac{1}{2}mv^2$$



E_k is kinetic energy measured in joules (J)
m is mass measured in kilograms (kg)
v is velocity measure in metres per second ($m.s^{-1}$)

A moving object has kinetic energy.

The amount of energy depends on the object's mass and on the square of its velocity.

Kinetic energy is a **scalar quantity**.

The kinetic energy of an object is **directly proportional** to the square of the magnitude of the velocity of the object, ie: $\mathbf{E_K \alpha v^2}$.

For example, if the magnitude of the velocity of the object is doubled, its kinetic energy will become $(2)^2$, thus **four times greater**.

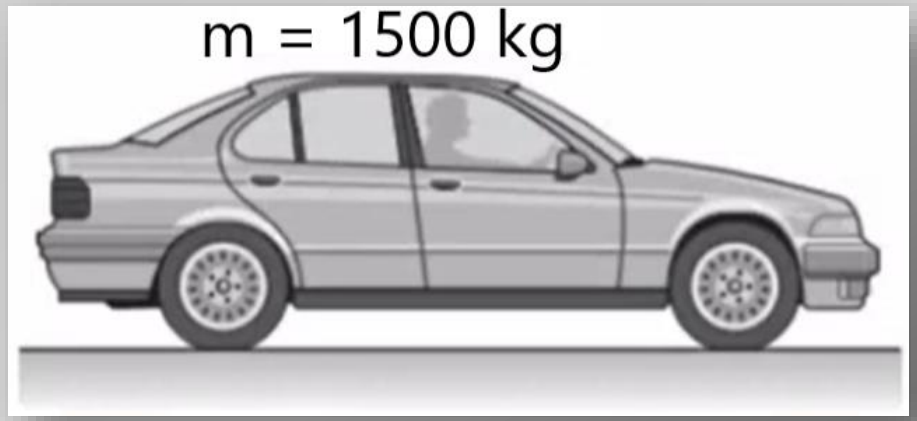
Example:

Calculate the kinetic energy of a car of mass 1500 kg when it travels at 72 $km.h^{-1}$.

Answer:

$72\ km.h^{-1} \div 3{,}6 = 20\ m.s^{-1}$

$$E_K = \tfrac{1}{2} mv^2$$
$$= \tfrac{1}{2} (1500)(20)^2$$
$$= 300\,000\ J$$



Question 1: A golf ball of mass 43,93 g, is struck off the tee at 58 $m.s^{-1}$. Calculate the kinetic energy of the golf ball as it leaves the tee.

$43{,}93\ g \div 1000 = 0{,}04393\ kg$

$E_K = \frac{1}{2} mv^2$
$= \frac{1}{2} (0{,}04393)(58)^2$
$= 73{,}89\ J$

Question 2: A soccer ball has 1200 J of kinetic energy and a mass of 0,42 kg. What is the speed of the ball when it is kicked?

$E_K = \frac{1}{2} mv^2$
$1200 = \frac{1}{2} (0{,}42).v^2$
$v = 75{,}59\ m.s^{-1}$

Question 3: A professional marathon runner runs at 21 $km.h^{-1}$ and possesses 1105,9 J of kinetic energy. What is the mass of the runner?

$21\ km.h^{-1} \div 3{,}6 = 5{,}83\ m.s^{-1}$

$E_K = \frac{1}{2} mv^2$
$1105{,}9 = \frac{1}{2} m.(5{,}83)^2$
$m = 65{,}07\ kg$

MECHANICAL ENERGY

Mechanical energy is: *the sum of gravitational potential and kinetic energy at a point*

Mathematically:

Mechanical energy = Gravitational Potential Energy + Kinetic energy

$$E_M = E_p + E_k$$

Mechanical energy is a scalar quantity and is measured in joules (J).

The mechanical energy of the object **at rest** in figure 1 is made up of gravitational potential energy (E_p) **only**.
The mechanical energy of the object in figure 2 is made up of gravitational potential energy (E_p) **and** kinetic energy (E_k).



Fig 1 Potential energy.

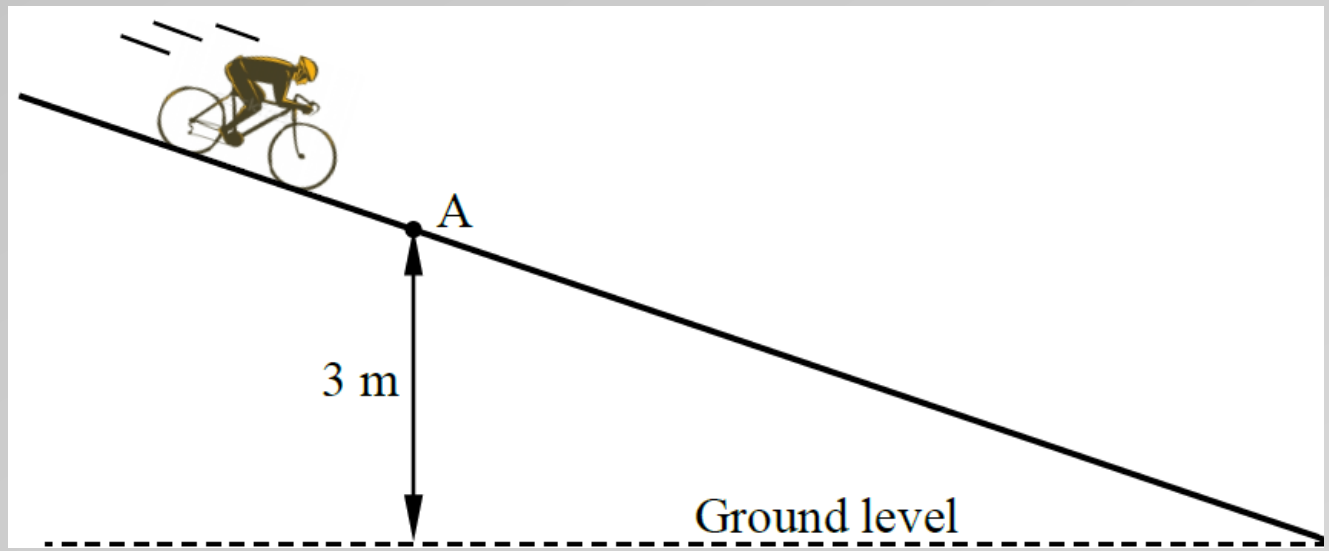


Fig 2 Potential energy and kinetic energy.

Example:

A cyclist is freewheeling down an inclined plane as shown below.

The speed of the cyclist is 10 $m.s^{-1}$ as she passes point **A** which is 3 m above the ground level. The total mass of the cyclist and her bike is 80 kg.



Calculate the mechanical energy of the cyclist and her bike at point **A**.

Answer:

$E_M = E_p + E_k$

$= mgh + \frac{1}{2}mv^2$

$= (80).(9,8).(3) + \frac{1}{2}(80).(10)^2$

$= 6352$ J

LAW OF CONSERVATION OF ENERGY

The law of conservation of energy: *the total energy in a system cannot be created nor destroyed, only transformed from one form to another*

THE PRINCIPLE OF CONSERVATION OF MECHANICAL ENERGY

The principle of conservation of mechanical energy: *In the absence of air resistance or any external forces, the mechanical energy of an object is constant*

$$E_{M\,(initial)} = E_{M\,(final)} \qquad (E_p + E_k)_i = (E_p + E_k)_f$$

Example 1: A falling object

When an object falls, it loses gravitational potential energy and gains kinetic energy. If there are **no frictional forces such as air resistance**, no work is done to overcome friction. Therefore, mechanical energy is conserved when no frictional forces are present.

The figure shows a falling ball. **At the highest point** of its motion the potential energy is a maximum and the **kinetic energy is zero**.

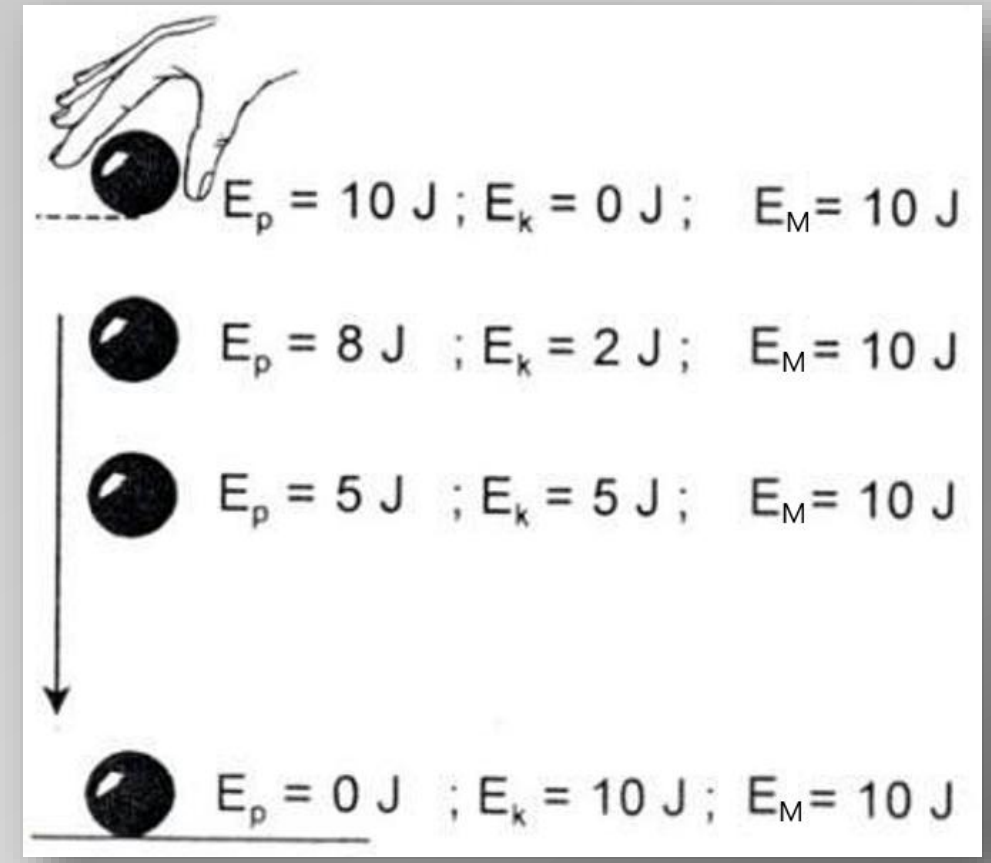
$E_M = E_p + E_k = 10 + 0 = 10\ J \quad \therefore E_M = E_p$

As the ball falls, potential energy is converted to kinetic energy.

As it **strikes the ground**, kinetic energy is a maximum and the **potential energy is zero**.

$E_M = E_p + E_k = 0 + 10 = 10\ J \quad \therefore E_M = E_k$

The total of the potential energy and kinetic energy however, remains the same.



Example 2: A simple pendulum

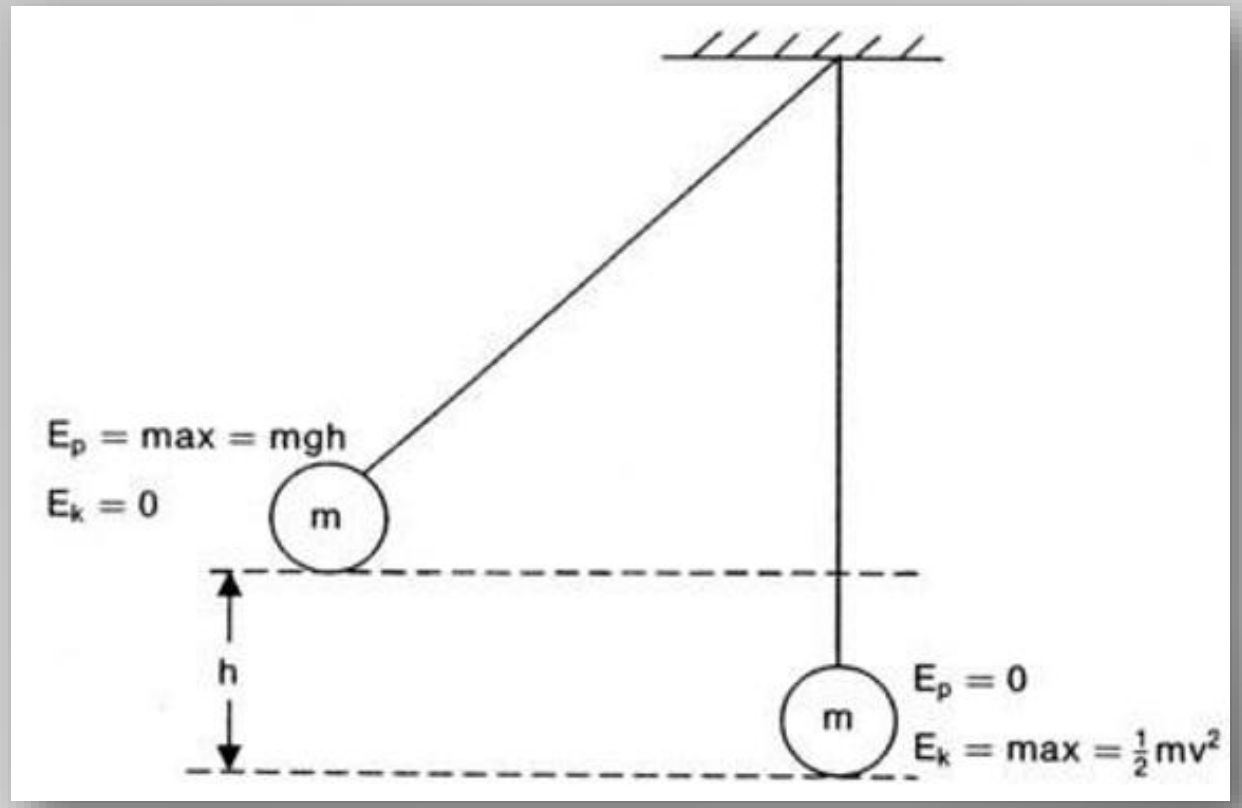
Mechanical energy is also conserved in the simple pendulum shown in the diagram. The pendulum begins to swing from a maximum height **h**.

$$E_{M\,(TOP)} = E_{M\,(BOTTOM)}$$

$$E_{p\,(TOP)} + E_{k\,(TOP)} = E_{p\,(BOTTOM)} + E_{k\,(BOTTOM)}$$

$$m.g.h + 0 = 0 + \tfrac{1}{2}\, m.v^2$$

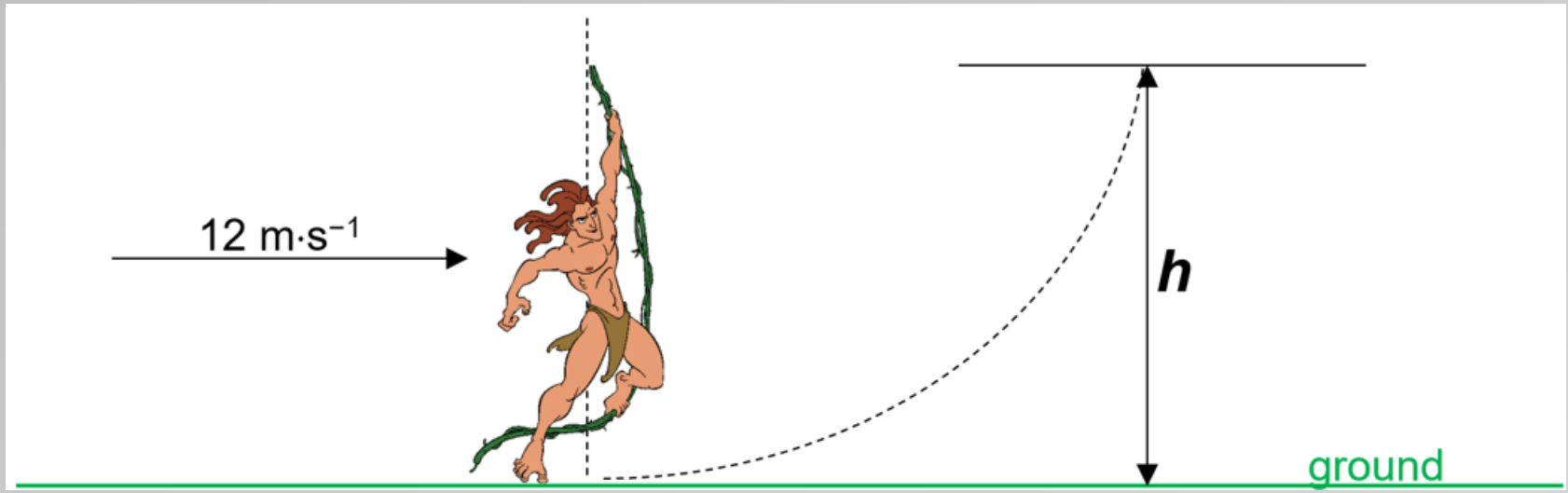
$$m.g.h = \tfrac{1}{2}\, m.v^2$$



Note:

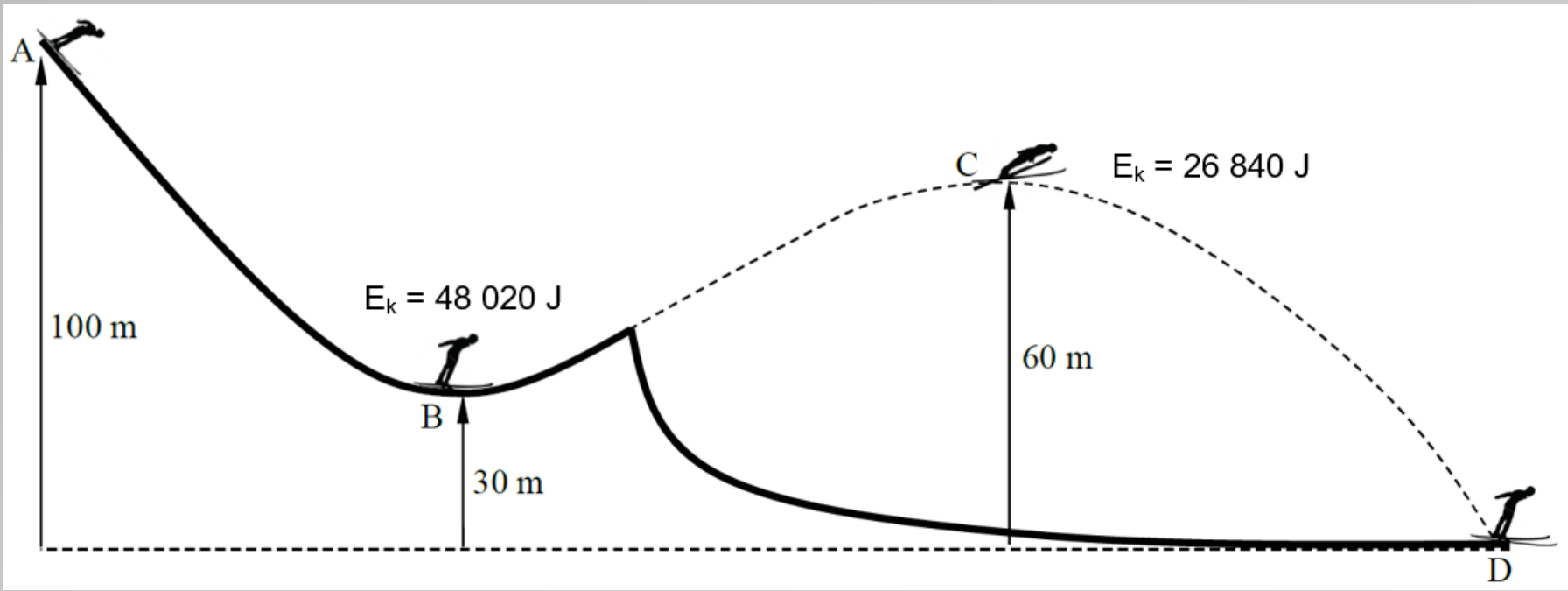
- Only pendulum problems may be started with the equation: $E_{p\,(TOP)} = E_{k\,(BOTTOM)}$
- Equations of motion may **not** be used, because it is not motion in a straight line.

Question 1: Tarzan runs at 12 $m \cdot s^{-1}$ along the ground and grabs onto a vine. Ignore air resistance. What maximum vertical height ***h*** will Tarzan reach above the ground?



$$
\begin{aligned}
E_{mech(ground)} &= E_{mech(h)} \\
0 + \tfrac{1}{2}m.(12)^2 &= m.(9{,}8).h + 0 \\
72.m &= 9{,}8.m.h \\
72 &= 9{,}8.h \\
h &= 7{,}35\ m
\end{aligned}
$$

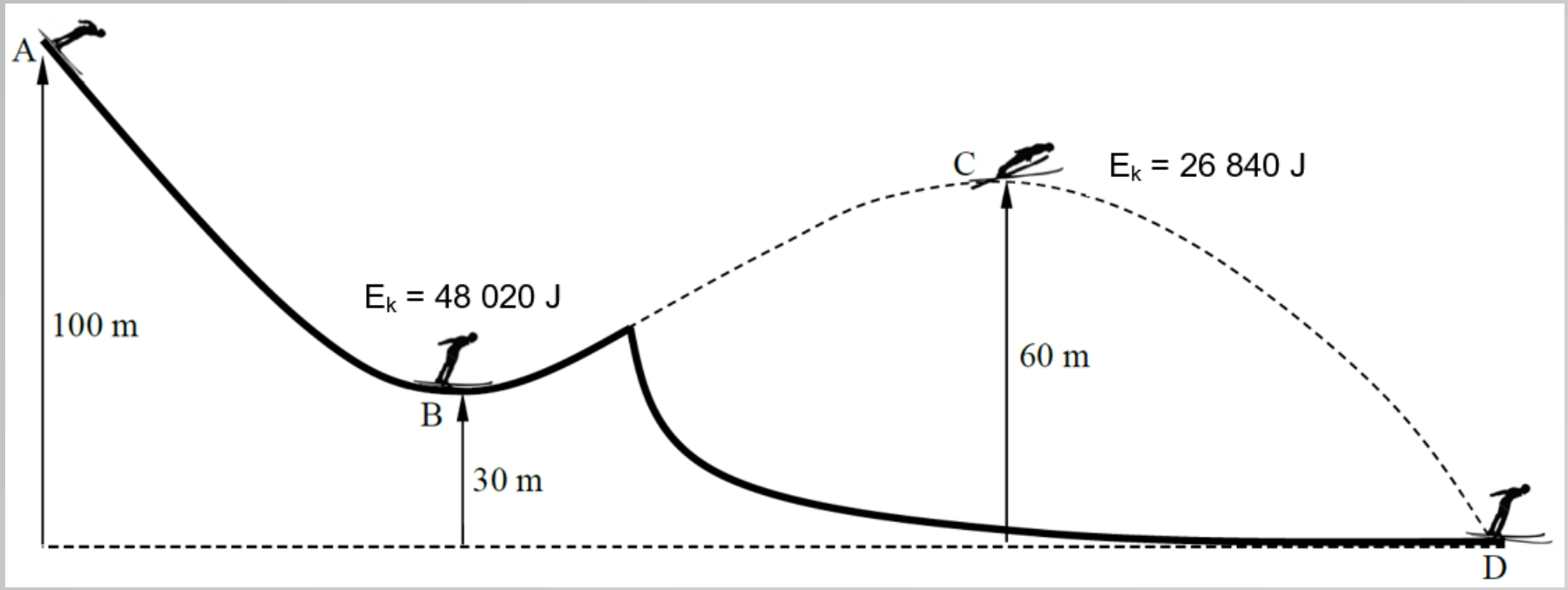
Question 2: A 70 kg skier starts from rest at point **A** and skis down a frictionless ramp eventually landing at point **D** as shown in the diagram below.



a) At which point would the skier possess the greatest gravitational potential energy relative to level **D**? at A

b) At which point would the skier have zero gravitational potential energy relative to level **D**? at D

A 70 kg skier



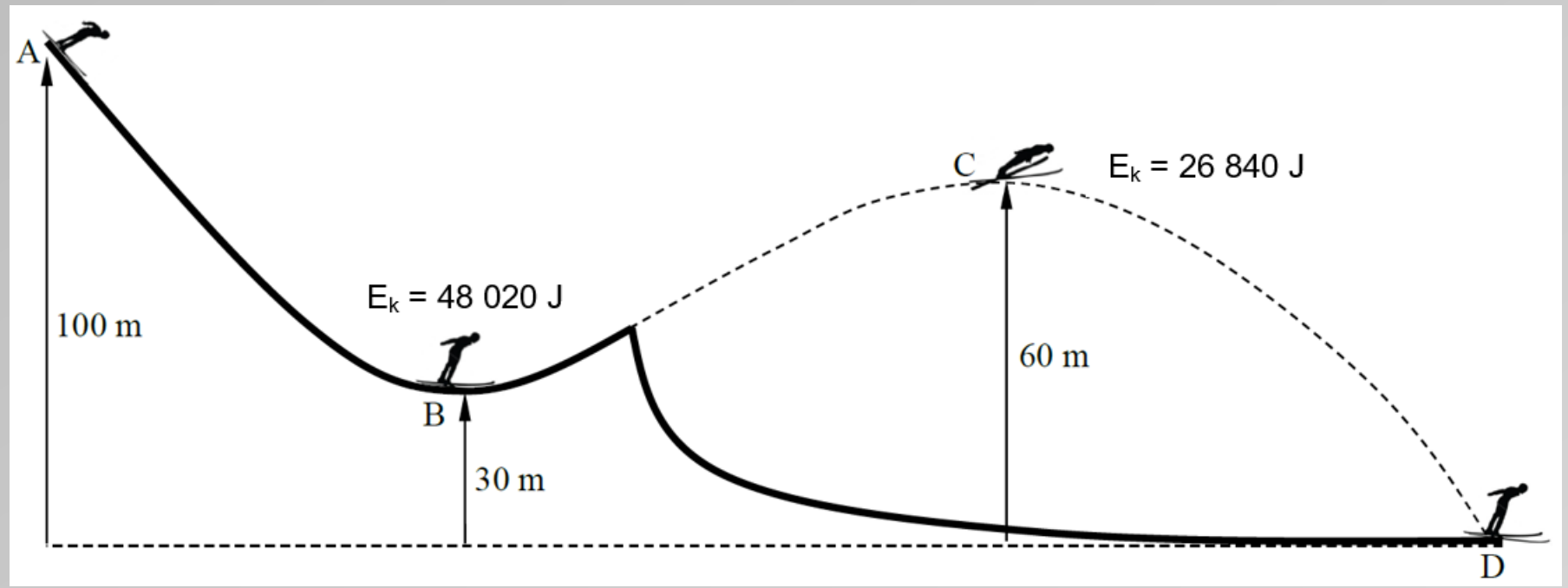
c) At what height above level **D** will the skier possess 20 000 J of gravitational potential energy?

E_p = m.g.h
20 000 = (70).(9,8).h
h = 29,15 m

d) Calculate the gravitational potential energy of the skier at point **A**.

E_p = m.g.h
= (70).(9,8).(100)
= 68600 J

A 70 kg skier



e) Calculate the speed of the skier at point **B**.

$E_k = \frac{1}{2}m.v^2$

$48020 = \frac{1}{2}(70).v^2$

$v = 37{,}04\ m \cdot s^{-1}$

f) Calculate the mechanical energy of the skier at point **C**.

$E_{mech} = E_p + E_k$

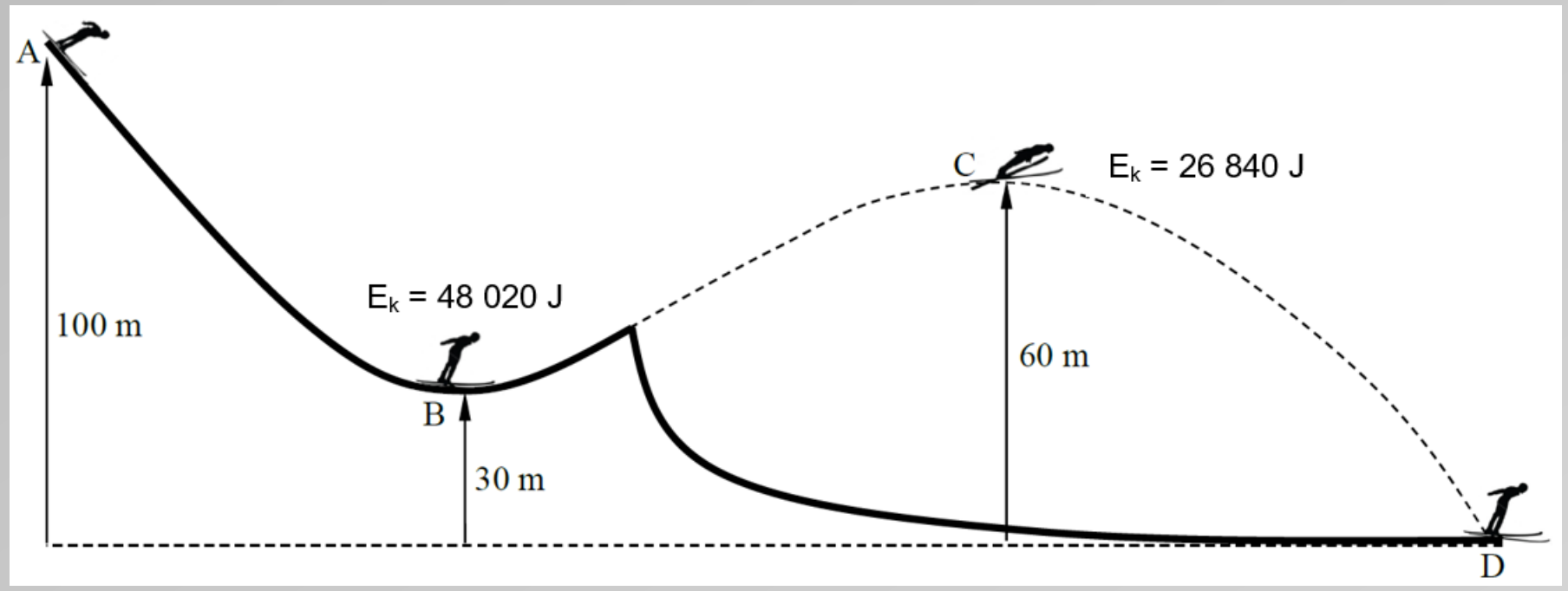
$= (70).(9{,}8).(60) + 26840$

$= 68000\ J$

A 70 kg skier

$E_{p\,(A)}$ = 68600 J

$E_{M\,(C)}$ = 68000 J



g) Has the mechanical energy of the skier been conserved as the skier moved from the ramp, through the air, to point **C**? Give a reason for your answer.

No.
Mechanical energy at **A** ≠ Mechanical energy at **C**

h) Has energy been conserved as the skier moved from the ramp, through the air, to point **C**? Give a reason for your answer.

Yes.
600 J of energy was converted to heat energy (removed from the skier by air friction) but the total energy of the system will remain constant.